Joao De Campos Carvalho

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EDUCATION

University of Toronto - BASc, Mechanical Engineering + PEY Co-op | Sep 2024 - Apr 2029
Minors: Robotics & Mechatronics, Engineering Business | GPA: 3.25 | CSWA-certified (Dec 2025)

EXPERIENCE

Auditorium AI - Operations Lead | Sep 2026 - present, Toronto

- Built the operating system — shared finance drive, per-event P&L tracker, deals pipeline, gear inventory, weekly founder reports — giving the founders a weekly view of money, deals, and gear.
- Manage a content lead against an approved brief with weekly check-ins, shipping 8 pieces per month on a steady cadence.
- Set up and managed an AI-assisted development workflow for the company web product, coordinating frontend-heavy AI-generated code through review into a working codebase.

Auditorium AI - Lead Design Engineer | May - Aug 2026, Toronto

- Engineered a spatial-calibration feature mapping automated stage lighting to room geometry, uncovering two silent failure modes that every existing check had passed.
- Integrated hardware and software for a single-board-computer field device — field updates, device driver rules, and a Fusion 360 3D-printed enclosure — delivering a complete unit ready for event deployment.
- Conducted market and competitive research that redirected the company's target customer segment, and produced the follow-up channel strategy.

UTESCA - VP of Outreach | Sep 2025 - present, Toronto

- Built an automated outreach tracker in Google Sheets and used it to identify partners at consulting firms, converting 80% of prospects into partners.
- Run a weekly prospecting cadence for event partnerships, contacting one consulting firm per week and tracking each from first touch to booked meeting.

UHN Sleep Monitoring Device - Co-Manager | Jan - Apr 2025

- Co-managed a 6-person team designing a low-cost sleep monitoring device with UHN, taking it from requirements to a working prototype.
- Originated 120+ design ideas and converged the team on a design 30% cheaper than industry competitors.

Brazilian Robotics Olympics - Team Leader / Chief Engineer | 2021 - 2023

- Designed and programmed the C# simulation code, placing top 5 nationally.

Robotics Project | Sep 2026 - present

- Contributing ~3 hrs/week to the mechanical build of an SO-101-based robotic arm, covering CAD, assembly, and testing.
- Own funding/sponsorship outreach and team admin/GitHub for a 3-person build team, running the team's sponsorship pipeline and build coordination.

PROJECTS

Eye-Tracker - Personal Project

- Building a webcam-based gaze-tracking control layer that delivers full hands-free computer control with zero keyboard or mouse input.
- Designed the gaze-plus-voice interaction model — open-palm gesture as push-to-talk, voice-only destructive confirmations, gaze/voice guided first-run tutorial — making hands-free control safe and learnable for first-time users.

SKILLS

Python, C++, PyTorch, TypeScript, MATLAB, Fusion 360, SolidWorks, Git, Linux, EasyEDA, Arduino, ESP32, Raspberry Pi

AWARDS

Bronze, Brazilian Mathematics Olympiad | Silver, Mathematique Sans Frontieres | People's Choice, U of T Undergraduate Engineering Research Day (UnERD 2026)